

Constructor Exercise

Exercise 1: Student Information

Create a class Student.

Variables:

String name;

int age;

double cgpa;

Create a constructor to initialize all variables.

Add a method:

void displayInfo()

Exercise 2: Rectangle

Create a class Rectangle.

Variables:

double length;

double width;

Create a constructor that initializes length and width.

Add these methods:

double calculateArea()

double calculatePerimeter()

void display()

Formula:

Area = length $\times$ width

Perimeter = 2 $\times$ (length + width)

Exercise 3: Employee Salary

Create a class Employee.

Variables:

String name;

double basicSalary;

double bonus;

Create a constructor to initialize the variables.

Methods:

double calculateTotalSalary()

void displaySalary()

Formula:

Total Salary = Basic Salary + Bonus

Exercise 4: Bank Account

Create a class BankAccount.

Variables:

String accountHolder;

double balance;

Create a constructor to initialize them.

Add methods:

void deposit(double amount)

void withdraw(double amount)

void checkBalance()

b1.checkBalance();

Make sure the user cannot withdraw more money than the current balance.

Exercise 5: Circle

Create a class Circle.

Variable:

double radius;

Create a constructor to initialize radius.

Methods:

double getArea()

double getCircumference()

void displayResult()

Exercise 6: Product

Create a class Product.

Variables:

String productName;

double price;

int quantity;

Create a constructor.

Add methods:

double calculateTotalPrice()

void displayProduct()

Formula:

Total Price = price × quantity

Exercise 7: Car

Create a class Car.

Variables:

String brand;

String model;

double price;

Create a constructor.

Methods:

void displayCar()

void applyDiscount(double percentage)

Exercise 8: Book

Create a class Book.

Variables:

String title;

String author;

double price;

Create a constructor.

Add methods:

void displayBook()

void discountPrice(double discount)

boolean isExpensive()

isExpensive() should return true if the book price is greater than 1000.

Exercise 9: Mobile Phone

Create a class Mobile.

Variables:

String brand;

String model;

double price;

int batteryLevel;

Create a constructor.

Methods:

void displayMobile()

void charge(int amount)

void useBattery(int amount)

Rules:

- Battery cannot exceed 100.
- Battery cannot go below 0.

Exercise 10: Constructor Overloading

Create a class Person.

Variables:

String name;

int age;

String city;

Create three constructors:

Person(String name)

Person(String name, int age)

Person(String name, int age, String city)

Add:

void display()

Exercise 11: Electricity Bill

Create a class ElectricityBill.

Variables:

String customerName;

int units;

Create a constructor.

Method

double calculateBill()

void displayBill()

Use these rules:

First 100 units = 5 taka/unit

Next 100 units = 7 taka/unit

Above 200 units = 10 taka/unit

Exercise 12: Student Result

Create a class StudentResult.

Variables:

String name;

double marks1;

double marks2;

double marks3;

Create a constructor.

Methods:

double calculateAverage()

String getGrade()

void displayResult()

Grade system:

80–100 = A+

70–79 = A

60–69 = A-

50–59 = B

40–49 = C

Below 40 = F

Exercise 13: Shopping Cart

Create a class ShoppingItem.

Variables:

String itemName;

double price;

int quantity;

Initialize them using a constructor.

Methods:

double getSubtotal()

double getDiscount()

double getFinalPrice()

void printBill()

Rules:

If subtotal $\geq 5000 \rightarrow 10\%$ discount

Otherwise $\rightarrow$ no discount

Exercise 14: Account with Constructor Validation

Create a class Account.

Variables:

String accountNumber;

String holderName;

double balance;

The constructor should make sure that the initial balance is not negative.

Methods:

void deposit(double amount)

boolean withdraw(double amount)

void displayAccount()

Exercise 15: Challenge — Library Book

Create a class LibraryBook.

Variables:

String title;

String author;

boolean borrowed;

The constructor should initialize:

borrowed = false;

Methods:

void borrowBook()

void returnBook()

void displayStatus()

Rules:

- A borrowed book cannot be borrowed again.
- A returned book cannot be returned again.

Exercise 16: Challenge — ATM Account

Create a class ATMAccount.

Variables:

String name;

int pin;

double balance;

Create a constructor.

Methods:

boolean verifyPin(int enteredPin)

void deposit(double amount)

void withdraw(double amount, int enteredPin)

void checkBalance(int enteredPin)

The withdrawal and balance-check methods should only work when the PIN is correct.